When the Judge Is Wrong: An LLM-as-Judge Reliability Benchmark Scored Against Ground Truth

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One-command reproducible · 5 judges × 39 ground-truth items + 29 ties + 12 self-preference questions

Abstract. "LLM-as-judge" — using a strong language model to grade the outputs of other models — is now the default evaluation method, but the judge is itself a fallible model with biases. Most studies measure a judge by its **agreement with humans** or with **other judges**; both are confounded, because raters and judges can share a bias and be wrong together. We instead measure judges against **ground truth**: a benchmark of items each with a known-correct and a plausibly-wrong answer, so a judge's accuracy can be scored directly, and its position, verbosity, self-consistency, calibration, and self-preference biases isolated as deviations from a perfect oracle. Across five frontier judges (GPT-4o, GPT-4o-mini, GPT-4.1, Claude Sonnet 4.6, Claude Haiku 4.5) we find a clean **dissociation**. On 39 objective items — including common misconceptions and counterintuitive-reasoning traps designed to induce error — judges are **near-perfect** (97–100% truth-accuracy), show **no position bias**, are **not fooled** by padding the wrong answer with authoritative filler, are perfectly self-consistent, and are well-calibrated. Yet on 29 **matched-quality** pairs, where both answers are fully correct and differ only in length, the same judges **strongly prefer the longer answer** (72–100%). A self-preference probe shows a modest own-family lean (+13 pt cross-family gap) once a length confound is controlled by differencing. The classic **position** bias appears solved; the classic **verbosity** bias is alive and strong, but surfaces only when quality is tied. The practical reading: LLM-as-judge is reliable for **verifiable** tasks and risky for **subjective** grading, where it rewards length over substance.

Keywords: LLM-as-judge · evaluation · ground truth · verbosity bias · position bias · self-preference · calibration · reliability · reproducibility

Contributions.

1. A **ground-truth** LLM-judge benchmark that scores accuracy against truth, not against humans or other judges, plus a position-robust accuracy metric.
2. A **dissociation** result across five frontier judges: near-perfect, unbiased judging on objective items versus strong verbosity bias on matched-quality ties.
3. A **self-preference** probe with a difference-in-differences design that controls a length confound, finding a modest +13 pt own-family lean.
4. An open, one-command, resumable harness with dated snapshots, released under MIT.

Contents

1 Introduction

Evaluation is the bottleneck of modern AI. As generative systems outrun the static benchmarks built to measure them, the field has converged on **LLM-as-judge**: prompt a strong model to grade the outputs of others [1], [2]. It is fast, cheap, and correlates well enough with human preference to power leaderboards [3]. But the judge is not an oracle; it is another fallible model, and a body of work has documented that it carries biases — a preference for the answer in a particular position [4], for longer answers [5], [6], and for its own generations [7]. If the instrument is biased, every measurement it produces inherits that bias.

How biased are **current** frontier judges, and where? This paper answers with a deliberately strict instrument. Most evaluations of judges measure **agreement with humans** or **inter-judge agreement**, but both are confounded: human raters carry their own biases (including the same length and position effects), and two judges can agree precisely because they share a bias. We avoid the confound by measuring against **ground truth**. Our items each have an objectively correct and a plausibly-wrong answer, so a judge's accuracy is scored directly against the truth, and its biases appear as departures from the behaviour of a perfect oracle.

The result is a clean **dissociation** that, we argue, resolves the apparent tension in the prior literature. On objective items — even ones engineered to trip a careless grader — five frontier judges are near-perfect and essentially unbiased: position bias is gone, and padding the wrong answer with confident, authoritative prose does not fool them. But the moment we remove the ground truth — pairing two answers that are **both correct** and differ only in length — the same judges revert to a strong preference for the longer one. The biases the literature reports are real, but they live where there is **no correct answer to anchor on**: in subjective, matched-quality comparisons, not in verifiable ones. That distinction is the contribution, and it has a direct practical consequence for anyone deploying an LLM judge.

1.1 Roadmap

Section 2 reviews LLM-as-judge and its known biases. Section 3 specifies the benchmark — ground-truth items, the position-robust truth-accuracy metric, and the tie and self-preference probes. Section 4 reports the three result blocks. Section 5 develops the dissociation and its implications. Sections 6–8 cover limitations, reproducibility, and future work.

2 Background and related work

2.1 LLM-as-judge and its biases

Using a model to evaluate models was popularized by MT-Bench and Chatbot Arena [1], [3] and is now surveyed as its own subfield [2]. The same work that established the method also flagged its failure modes: **position bias**, the tendency to favour whichever answer appears first (or second), shown to be large enough to flip verdicts by reordering [4]; **verbosity (length) bias**, the tendency to prefer longer answers independent of quality [5], severe enough that leaderboards now apply explicit length control [6]; and **self-enhancement / self-preference**, the tendency of a judge to favour text from its own model, which has been linked to the model's ability to **recognize** its own generations [7]. Our benchmark measures all three on current frontier models, with a design that separates genuine bias from the length and quality confounds that complicate the self-preference case.

2.2 Why ground truth, not agreement

The dominant validation for an LLM judge is correlation with human ratings. This is necessary for subjective tasks but is a weak test of **reliability**, because human labels carry the very biases under study — annotators also prefer longer, more fluent answers — so a judge can "agree with humans" by sharing their bias. Inter-judge agreement is weaker still. We therefore restrict to items with an **objective** answer (verifiable facts, arithmetic, logical entailment, code behaviour), where truth is independent of any rater, and treat agreement-based evaluation as complementary rather than primary.

2.3 Scoring and calibration

We score binary judge decisions against truth and summarize calibration with the Brier score [8], a strictly proper scoring rule [9], using the judge's self-reported confidence. Edit distance [10] underlies an auxiliary check. One item class — counterintuitive reasoning such as the bat-and-ball problem — is drawn from the cognitive-reflection literature [11], chosen because the **intuitive** answer is wrong, which tests whether a judge is seduced by a fluent wrong answer.

3 Method

3.1 Items and ground truth

The core set is 39 items, each a triple `(question, correct, wrong)` with an objectively correct answer and a **plausibly** wrong distractor (subtle, not absurd, so that trivial rejection does not inflate scores). Items span facts, arithmetic, logic, and code (24 "easy"), plus 15 "hard" items chosen to **induce** error: common misconceptions where the wrong answer is the popular belief (blood in veins is blue; we use 10% of our brain; the tongue taste-map), and counterintuitive reasoning where the wrong answer is the intuitive one (bat-and-ball, Monty Hall, the doubling lily pads). The hard set is the real test — if a judge merely parrots common belief, it fails here.

3.2 Position-robust truth-accuracy

For a pairwise item the judge is asked, in a fixed minimal rubric, which answer is better, in **both** answer orders. It is **truth-correct** on the item only if it picks the correct answer in **both** orders:

$$\text{truth-correct}_r = \mathbb{1}[J(\text{correct}, \text{wrong}) = \text{correct}] \cdot \mathbb{1}[J(\text{wrong}, \text{correct}) = \text{correct}], \tag{1}$$

so a coin-flipper or a pure position-follower is penalized. We report truth-accuracy (the headline), naive per-order accuracy (for contrast), first-slot rate (50% = unbiased), order-consistency, a verbosity-flip rate (truth-correct in plain but not when the wrong answer is padded with authoritative filler), self-consistency over $K\{=\}3$ repeats at non-zero temperature, and the Brier score of the judge's confidence.

3.3 The tie probe (matched quality)

The sensitive test of verbosity bias removes the ground truth. Each of 29 **tie** items pairs two answers that are **both fully correct** and differ only in length — a terse correct answer and a longer correct answer padded with true, relevant elaboration. A perfect judge should be indifferent (50/50). We report **long-preference**: the fraction of judgments (over both orders) that pick the longer answer. Above 50% is verbosity bias, measured on genuinely equal quality.

3.4 The self-preference probe (difference-in-differences)

For 12 open-ended questions with no single right answer, we generate one answer from an OpenAI-family model and one from an Anthropic-family model (each constrained to one concise sentence), and have every judge pick the better, in both orders. The **absolute** own-family preference is confounded by answer quality and length; we therefore report the **cross-family gap** — the difference between how often OpenAI judges and Anthropic judges prefer the OpenAI answer. Because every judge sees the **identical** answer pair, any shared property (including length) cancels in this difference, leaving a length-controlled estimate of self-preference.

3.5 Judges and implementation

Five judges: GPT-4o, GPT-4o-mini, GPT-4.1 (OpenAI) and Claude Sonnet 4.6, Claude Haiku 4.5 (Anthropic). Deterministic probes run at temperature 0; the self-consistency probe at temperature 1. All calls cache to disk and the harness resumes, so a run retries only missing cells. The roster, items, ties, and questions are plain JSON and trivially extensible. (Claude Opus 4.8 was excluded: it rejects the temperature control the method requires.)

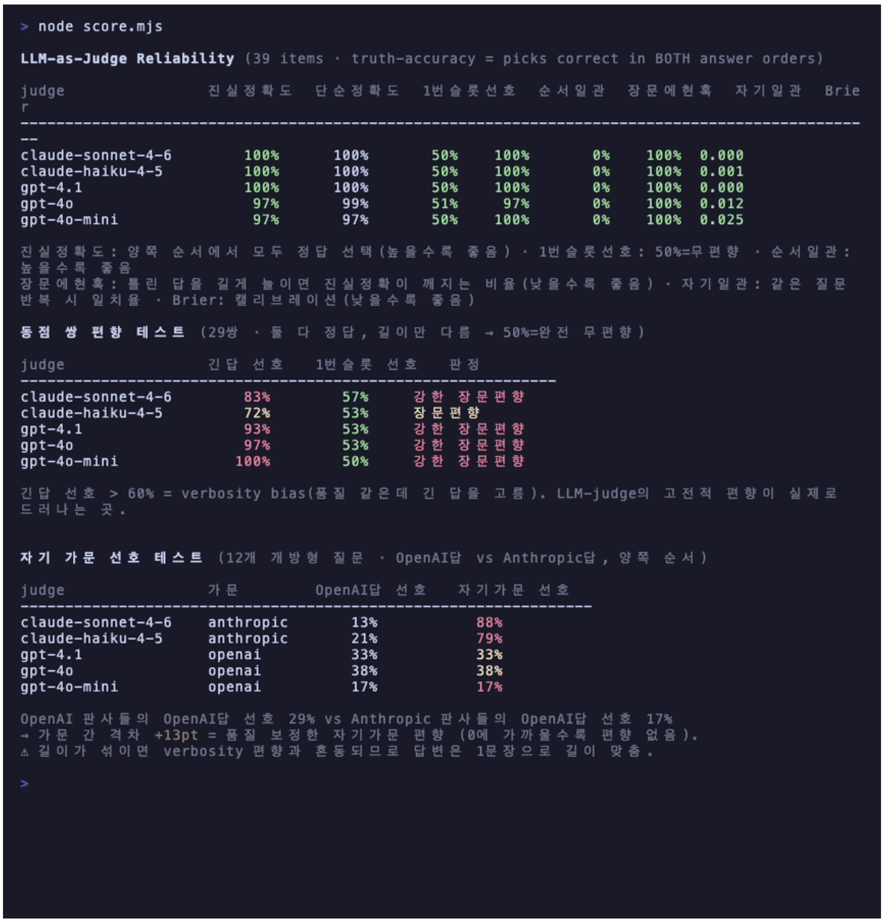

Figure 1: The harness prints all three result blocks — ground-truth reliability (top), the matched-quality tie test (middle), and self-preference (bottom) — reproducible with one command.

4 Results

4.1 On objective items, judges are near-perfect and unbiased

Table Table 1 reports the ground-truth block. All five judges score **97–100%** truth-accuracy — including on the hard misconception and reasoning items — with **first-slot rates at 50%** (no position bias), **0% verbosity-flips** (padding the wrong answer never fools them), **perfect self-consistency**, and **near-zero Brier** (well-calibrated confidence). The classic position bias, large in earlier studies [4], is simply absent here, and a fluent, authoritative wrong answer does not survive contact with a correct one.

Judge	Truth-acc	First-slot	Order-cons.	Verb-flip	Self-cons.	Brier
claude-sonnet-4-6	**100%**	50%	100%	0%	100%	0.000
claude-haiku-4-5	**100%**	50%	100%	0%	100%	0.001
gpt-4.1	**100%**	50%	100%	0%	100%	0.000
gpt-4o	97%	51%	97%	0%	100%	0.012
gpt-4o-mini	97%	50%	100%	0%	100%	0.025

Table 1: Ground-truth reliability over 39 objective items (24 easy + 15 hard misconception/reasoning). Truth-accuracy requires picking the correct answer in both orders.

4.2 On matched-quality ties, the same judges reward length

Remove the ground truth and the picture inverts (Table Table 2). On 29 pairs where both answers are correct and differ only in length, every judge prefers the **longer** answer far above the unbiased 50%: gpt-4o-mini 100%, gpt-4o 97%, gpt-4.1 93%, Claude Sonnet 83%, Claude Haiku 72%. Position preference on these ties stays near 50–57%, confirming the effect is length, not order. The verbosity bias the literature reports [5], [6] is not only present but strong — it was merely invisible on the objective items, where correctness dominated the decision.

Judge	Prefers longer	First-slot	Verdict
gpt-4o-mini	**100%**	50%	strong verbosity bias
gpt-4o	**97%**	53%	strong verbosity bias
gpt-4.1	**93%**	53%	strong verbosity bias
claude-sonnet-4-6	**83%**	57%	strong verbosity bias
claude-haiku-4-5	**72%**	53%	verbosity bias

Table 2: Matched-quality tie test: 29 pairs, both answers fully correct, differing only in length. 50% = unbiased; every judge prefers the longer answer.

4.3 A modest self-preference survives length control

On the open-ended questions, every judge preferred the Anthropic-generated answers in absolute terms — but those answers were on average longer (158 vs 131 characters), so given the verbosity bias just measured, the absolute numbers are a length artifact, not loyalty. The clean signal is the **cross-family gap**: OpenAI judges prefer the OpenAI answer 29% of the time versus 17% for Anthropic judges, a **+13 pt** lean toward their own family despite those answers being shorter (Table Table 3). Because all judges see the identical pair, length cancels in this difference. The self-preference reported elsewhere [7] is therefore present but modest here, and easy to mistake for — or have masked by — verbosity bias.

Judge	Family	Prefers OpenAI ans.	Prefers own family
claude-sonnet-4-6	anthropic	13%	88%
claude-haiku-4-5	anthropic	21%	79%
gpt-4.1	openai	33%	33%
gpt-4o	openai	38%	38%
gpt-4o-mini	openai	17%	17%

Table 3: Self-preference over 12 open questions. Absolute own-preference is length-confounded; the length-controlled signal is the cross-family gap (OpenAI judges 29% vs Anthropic judges 17% prefer the OpenAI answer → +13 pt).

5 Discussion

5.1 The dissociation

The headline is a dissociation: **where there is a correct answer, frontier judges find it and are not swayed by length or order; where quality is tied, the same judges systematically reward length.** This reconciles two narratives. The "LLM judges are biased" literature [4], [5] is right that the biases exist; our result adds **where** — they are gated by the presence of ground truth. When correctness can decide the comparison, it dominates, and the biases vanish into the noise; when it cannot, the biases drive the verdict. Position bias, notably, appears to have been **engineered away** in current models (≈50% first-slot everywhere), while verbosity bias has not.

5.2 Implications for using LLM judges

The practical rule follows immediately. **Use LLM-as-judge freely for verifiable tasks** — factual correctness, unit-test pass/fail, exact-match — where our objective-item reliability is excellent. **Distrust it for open-ended, subjective grading** — essays, helpfulness, "which response is better" — where the tie result shows it will reward length over substance, inflating verbose answers regardless of quality. Where subjective judging is unavoidable, **control length explicitly** [6], randomize position (already largely safe), and prefer panels across model families to dilute the residual self-preference. The single most actionable finding is that a verbose answer enjoys a large, systematic advantage before a single point of substance is considered.

5.3 Honesty about confounds

We are explicit that the self-preference probe is confounded by answer length and quality, and we report only the difference-in-differences gap as the controlled estimate; the absolute own-preference numbers are shown precisely so the confound is visible rather than hidden. This is the same discipline the dissociation itself enforces: the bias is real but conditional, and reporting the condition is the result.

6 Limitations and threats to validity

Curated items, small n. 39 objective items, 29 ties, 12 self-preference questions, five models, one dated snapshot. The effects (verbosity 72–100%, truth-accuracy 97–100%) are large relative to n, but the benchmark is a **snapshot**, not a verdict, and the fine ordering among near-perfect judges is not meaningful. Correct/wrong labels and tie-equality are author judgments.

Objective scope. By design the ground-truth method covers only tasks with an objective answer; it cannot certify judge reliability on genuinely subjective quality, which is exactly where the tie result warns the danger lies.

Self-reported confidence. Calibration uses the judge's stated confidence, itself a model output.

Self-preference confound. Generated answers are not length-matched; we mitigate by differencing, but the gap rests on n = 12 and one model per family.

Two providers. Judges and answer-generators span OpenAI and Anthropic only; other families (and open models) would strengthen the self-preference design.

7 Reproducibility

The harness is open under the MIT license and runs with one command given an OpenAI and an Anthropic key (`node run.mjs`), printing the leaderboard with `node score.mjs`; an offline `--mock` judge validates the metric logic without any API cost. All calls cache to `results.json` and the run resumes, so a partial run completes cheaply; the full snapshot is a few dollars. Items, ties, and questions are plain JSON; the raw verdicts and the dated `REPORT.md` are released so every number is checkable and the analysis re-runnable.

8 Future work

The decisive extensions are **scale and breadth**: more items on a controlled difficulty grid with seed sweeps to turn the snapshot into interval estimates; more judges, including open models and a third provider, to sharpen the self-preference design; and **length-matched** generated answers (or post-hoc length normalization) to estimate self-preference without leaning on differencing. A **subjective-task** companion — where ground truth is replaced by careful human adjudication on length-controlled pairs — would extend the reliability map into the region this benchmark deliberately cannot reach. Finally, the tie probe suggests an intervention worth testing directly: whether instructing or fine-tuning judges to ignore length closes the 72–100% gap without harming objective accuracy.

9 Conclusion

LLM-as-judge is the instrument the field now measures itself with, so the instrument's reliability is foundational. Scoring five frontier judges against ground truth, we find they are near-perfect and unbiased where a correct answer exists — position bias appears solved, and authoritative wrong answers do not fool them — yet strongly biased toward length the moment quality is tied, with a modest residual self-preference. The lesson is not "LLM judges are reliable" nor "LLM judges are biased," but the conjunction: **reliable where verifiable, biased where**

subjective. Knowing which regime you are in is the difference between a trustworthy measurement and a confident mistake.

Data and code availability. The harness, items, ties, open questions, raw verdicts (`results.json`), and dated `REPORT.md`, plus this paper's source, are open under MIT and reproducible with one command.

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Appendix A — Sample items

Representative items (full sets in the repository's `data/`). Each objective item has a correct and a plausibly-wrong answer; the hard set targets misconceptions and counterintuitive reasoning.

Type	Correct	Plausibly wrong
misconception	Human blood is always red; veins look blue through skin.	Deoxygenated blood in veins is blue.
misconception	We use virtually all of the brain.	Humans use only 10% of their brains.
reasoning (CRT)	The ball costs $0.05.	The ball costs $0.10.
reasoning	Switching wins the Monty Hall game with prob. 2/3.	It is 50/50, so switching makes no difference.
code	`0.1 + 0.2 === 0.3` is false (floating point).	It is true.
fact	On Venus a day is longer than a year.	A year is longer, as on Earth.

Appendix B — Metric definitions

Truth-accuracy requires the correct pick in both answer orders (Section 3.2). **Long-preference** (ties) is the fraction of tie-judgments choosing the longer of two equally-correct answers; 50% is unbiased. **Self-preference gap** is $p_{\text{OpenAI judges prefer OpenAI}} - p_{\text{Anthropic judges prefer OpenAI}}$, a difference-in-differences that cancels any property shared by the fixed answer pair, including length. **Brier** $= \frac{1}{N} \sum (c_i - y_i)^2$ over judgments, where c_i is the judge's confidence (0–1) that its pick is correct and y_i whether it was; lower is better, 0.25 is no-skill.